

Unit 7 Assignment: Google Cloud Responsible AI

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Unit 7 Assignment Answer sheet

Question 1

I believe responsible AI should be integrated into healthcare culture through structured training, leadership expectations, and formal review processes. This will ensure safe practices before AI tools are used in patient care. Healthcare organizations need to treat AI like any other clinical system. They need to emphasize practices where safety, accuracy, and accountability are built into everyday workflows rather than added after deployment. A major challenge is that many clinicians are still skeptical of AI, especially when it comes to trusting automated decisions that impact patient outcomes. However, this skepticism also creates an opportunity to build a stronger AI-ready workforce that understands both the benefits and limitations of these systems. Research shows that successful AI adoption depends heavily on change management and workforce readiness, not just the technology itself (Anderson, 2024), which reinforces the need to align culture with responsible AI practices.

Question 2

Some of the most critical ethical concerns in healthcare are wrong predictions, privacy, bias, and over-reliance. Though, I think wrong predictions are the most serious because they can directly impact patient survival. If an AI system produces an incorrect diagnosis or treatment recommendation, the consequences are not just operational, but potentially life threatening. Privacy is also a major concern because AI systems rely on large volumes of sensitive patient data. This data must be protected to maintain trust. Transparency plays an important role because both providers and patients need to understand how decisions are made, especially in life or death scenarios. As IBM states, "For the public to trust AI, it must be transparent" (IBM, 2024), which highlights that ethical AI is not only about performance, but also about building trust and accountability.

Question 3

I think responsible AI will transform healthcare jobs by creating new oversight focused roles rather than replacing workers. Instead of eliminating positions, AI will shift responsibilities toward reviewing outputs, validating decisions, and ensuring systems are functioning correctly. This means healthcare professionals will need new skills. They will need to learn new tasks such as understanding AI limitations, identifying errors, and knowing when to override automated recommendations. Organizations will need to invest in workforce development programs that combine technical training with ethical

awareness. Over time, this shift will create a workforce that is better equipped to collaborate with AI systems while maintaining control over critical decisions.

Question 4

A key use case where fairness and bias could be especially problematic is AI-assisted diagnosis. This is particularly true when systems are trained on non-representative data. If a model is primarily trained on one population, it may perform well for that group but produce inaccurate results for others. This can lead to unequal care. This creates serious ethical concerns because healthcare decisions must be fair and consistent across all patients. To address this, organizations must use diverse datasets, continuously test for bias, and include human oversight in all high risk decisions. According to NIST, trustworthy AI should be “valid and reliable, safe, secure and resilient, accountable and transparent, explainable and interpretable, privacy-enhanced, and fair with harmful bias managed” (NIST, n.d.), which shows how many layers are required to ensure fairness in practice.

Question 5

Over the next decade I believe responsible AI will significantly improve healthcare by increasing efficiency, supporting better decisions, and improving patient outcomes. I believe AI will primarily assist workers rather than replace them. This is especially true because healthcare relies heavily on human judgment and context. However, this will require organizations to build a culture that prioritizes governance, oversight, and continuous learning alongside innovation. An academic perspective supports this idea, noting that “fairness and equity must be inherent to the design of medical AI” (Stone, 2024), which emphasizes that ethical considerations must be built into systems from the beginning. The biggest benefit will be stronger decision support for clinicians. The biggest challenge will be maintaining trust, accountability, and ethical standards as AI becomes more embedded in healthcare systems.

References

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